Journal of Sport Rehabilitation, (Ahead of Print)
https://doi.org/10.1123/jsr.2022-0076

First Published Online: Aug. 17, 2022

Human Kinetics
TECHNICAL REPORT

Variability in Normalization Methods of Surface Electromyography Signals in Eccentric Hamstring Contraction

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Context: In human movement analysis, normalization of a surface electromyography signal is a crucial step; therefore, parameter selection for this procedure must be adequately justified. The aim of this research was to determine the variability of electromyography signals in eccentric hamstring contraction under different normalization parameters. ***Design:*** Cross-sectional study. ***Methods:*** Nine university rugby players (age 21.50 [3.61] y; body mass index 21.50 [4.95]) and no history of recent hamstring injury. Values from maximum voluntary isometric contraction protocol and task related (ie, Nordic hamstring exercise) were used for surface electromyography signal normalization. Intersubject and intrasubject variation coefficients were used for normalization method variability and for signal reproducibility, respectively. ***Results:*** Intrasubject variation coefficient value indicates acceptable reproducibility of surface electromyography (less than 12%) for all normalization procedures. Lower values of intersubject variation coefficient value were achieved for normalization procedures using task-related values. ***Conclusion:*** Parameters extracted from task execution provided less variability for surface electromyography amplitude normalization in eccentric hamstring muscle contractions.

Keywords: muscle contraction, college athletes, lower-extremity

Surface electromyography (sEMG) is a technique used for recording signals associated with electrical activity from skeletal muscles. One of the most common applications of sEMG is the estimation of the activation and relaxation of muscles during different phases of a movement.[1] In this analysis, technical and physiological factors that may influence the amplitude of the recorded signals must be taken into consideration to enable comparisons between subjects and muscles.

Comparing muscle activity during a dynamic task can be very useful in both clinical and research environments; however, the changes in joint angles and force can be a source of many confounding factors of sEMG recordings. In this context, signal amplitude description may be highly variable and therefore is recommended to normalize the amplitude and express muscle activation as a percentage of a reference value.[2,3]

Among the commonly used parameters to normalize are the maximum signal amplitude in an isometric voluntary contraction protocol, maximum peak in the execution of the task used in experimental protocol,[4,5] or an average of the maximum activation signal bursts of maximum voluntary isometric contraction (MVIC)- or task-related peaks.[6]

As normalization of a sEMG signal is a crucial step in dynamic contractions assessment, parameter selection for this procedure must consider the type of muscle contraction. However, there is no consensus on which amplitude normalization method is better for enabling comparisons between subjects during eccentric muscle regimes, such as in the Nordic hamstring exercise. Therefore, the aim of this research was to determine the variability of electromyography signals in eccentric hamstring muscle contraction under different normalization parameters.

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Methods

Study Design

This study was conducted using a cross-sectional design. We defined sEMG signals' variability (ie, between-subject variability) as dependent variable and normalization procedures as independent variables.

Participants

Given that, among sports involving kicking and running, the most frequent injuries are noncontact muscle/tendon injuries,[7] athletes who were part of the rugby team of the University of Antofagasta were invited to participate voluntarily in this research. The inclusion criteria considered were practicing the discipline at least 3 times a week and having 1 competition match per week, age between 18 and 25 years (21.50 [3.61] y), body mass index < 35 (21.50 [4.95]), and no history of hamstring injury in the last 6 months. Nine players met the inclusion criteria and signed an informed consent form. The research protocol for this research was in accordance with the Declaration of Helsinki (1975, subsequently revised in 2008).

Procedures

sEMG Measurement. The sEMG for the noninvasive assessment of muscles recommendations were followed for sensor placement[8]. The skin surface on which sensors were placed was shaved and cleaned with an alcohol solution. Subsequently, Ag/AgCl surface hex-shaped electrodes (adhesive area of 4 × 2.2 cm, diameter of 1 cm, interelectrode distance of 2 cm, Noraxon Corporate) were positioned bilaterally, on biceps femoris (right and left) and semitendinosus (right and left) muscles. For electrode positioning, participants were asked to remain prone on a procedure table. For biceps femoris, electrodes were positioned at 50% of a line joining the ischial tuberosity and the lateral epicondyle of the tibia, while for semitendinosus electrodes were positioned at 50% of a line between the ischial tuberosity and the medial epicondyle of the tibia. The sEMG signals were recorded telemetrically using Noraxon TeleMyo DTS Desk Receiver (500 μV amplifier, common rejection ratio >100 dB, baseline noise <1 μV RMS, input impedance >100 MΩ, 16-bit resolution analog-to-digital card, Noraxon Corporate), at a sampling rate of 1500 Hz. Signals were displayed on the MyoResearch (MR) platform (version 3.8, Noraxon Corporate).

Maximal Voluntary Isometric Contraction Protocol. Participants performed a warm-up on a stationary bicycle for 5 minutes. For MVIC recording, the make test was performed[9]. Participants were positioned in prone position on a clinical table. The knee was maintained in 90° of flexion using a digital goniometer (DTS dual axis goniometer, Noraxon Corporate). Participants were asked to perform a knee flexion for a period of 5 seconds, while the evaluator manually resisted this position (ie, "make test") using the online goniometer recording as feedback. The instructions given by the evaluator were "push as hard as you can." The evaluator also provided verbal motivation to improve performance on the MVIC. This procedure was repeated 3 times, with 1 minute of rest between sets.

Eccentric Exercise. To record eccentric contraction of biceps femoris and semitendinosus, the Nordic hamstring exercise (NHE) was used. All participants were shown a video of NHE. The NHE was performed on an EN-Tree Train bench (Enraf-Nonius), which has a proximal leg restraint system to stabilize the lower-extremity during NHE execution. In addition, a metronome was used to control the descent during NHE execution with a frequency of 30 beats per minute to ensure that each repetition was performed in at least 3 seconds. Five repetitions of the NHE were recorded, with 2 minutes of rest between repetitions to avoid muscle fatigue[10].

Signal Processing. Signal processing was performed in MATLAB (R2017b). Raw electromyography (EMG) signals (ie, MVIC and NHE repetitions) were filtered using a fourth-order band pass Butterworth filter with 20 and 450 Hz as cutoff frequencies. Then, signals were smoothed using a nonoverlapping moving average window with a 100-ms time window[11].

Normalization. To normalize the sEMG signals, 2 values extracted from the maximum voluntary contraction protocol and 2 values from the repetitions were considered:

- Maximum value of the MVIC test (mMVIC): The highest value achieved among the 3 repetitions of MVIC was selected.
- Average of MVIC values: The maximum values of each of the 3 mMVICs were selected and averaged to generate a single value.
- Maximum eccentric activity value (mP): The maximum value of each NHE repetition is considered. Subsequently, each repetition was normalized to its own maximum reached.
- Average of maximum values of eccentric activity (aP): The maximum values of each repetition of NHE per subject were extracted. Subsequently, the maximum values were averaged, and the signals were normalized with this obtained value.

After normalization, the electromyographic signal data between the onset of activity and the maximum peak of the recording were extracted. To determine the onset of electromyographic activity, a threshold corresponding to the mean of a window from the first 150 samples (ie, without electromyographic activity) plus 3 SDs of each signal were used.[12] Then, signals were resampled to 1000 samples to adjust all signals to the same size. As the electromyographic signal of an eccentric contraction does not present a constant behavior (ie, increases over time), signals were divided in 3 temporal ranges of electromyographic activity (R1: 0%–33%, R2: 33%–66%, and R3: 66%–100%) for a detailed description of the signal.[13] From each range, the mean and SD were extracted to determine the variability of the electromyograms.

Statistical Analysis

To determine the variability between electromyograms, according to the normalization method, the intersubject (inter-CV) and intrasubject (intra-CV) variation coefficients were used. The inter-CV corresponds to the division of the SD of the series by the mean of the same series (ie, each repetition) of each normalization method.

To estimate the intra-CV, first the total mean squared error was calculated using repeated-measures analysis of variance. Then, the intra-CV corresponds to the quotient between the square root of the total mean square error and the mean electromyographic activity normalized by method. It has been reported that an intra-CV greater than 20% is indicative of high variability in the sEMG values and a value less than 12% indicates low variability and therefore an acceptable replicability of the signals.[14]

For the estimation of intra-CV and inter-CV values, the 5 NHE repetitions were considered. The Shapiro–Wilk test was used to determine the distribution of intra-CV and inter-CV. Data were presented as mean (SD) in case of normal distribution and as median and 25% and 75% percentiles in case of nonnormal distribution. Statistical analysis was performed in SPSS (version 25), and an $\alpha = .05$ was considered as the tolerance level.

Results

Intrasubject Variation Coefficients

Table 1 shows the intrasubject coefficients of variation. None of the intra-CVs exceeds 20%, so the criteria for reproducibility and stability of electromyography signals are met.

Intersubject Variation Coefficients

Consistently, lower inter-CV values are observed when using task-related values (ie, mP and aP) as normalization parameters, compared with values extracted from the MVIC tests (ie, mMVIC and average of MVIC) (Figures 1–4).

Discussion

The aim of this research was to determine the variability of electromyography signals in eccentric hamstring contraction under different normalization parameters using values extracted from an MVIC protocol and from 5 NHE repetitions. Our main results suggest that normalization methods using the averages of the maximum peak activity of repetitions (aP) and the maximum per repetition (mP) achieve lower inter-CVs compared with

Table 1 Intrasubject Variation Coefficients by Range (ie, R1: 0%–33%, R2: 33%–66%, and R3: 66%–100%) in rBF and lBF and rST and lST by Normalization Method

		Normalization method			
Ch	R	mP	aP	mMVIC	aMVIC
rBF	R1	8.31 [3.28; 14.5]	7.73 [3.26; 15.27]	6.42 [2.89; 14.61]	7.59 [3.36; 17.14]
	R2	3.13 [1.25; 6.72]	3.40 [1.26; 6.32]	3.07 [1.07; 7.77]	3.45 [1.23; 9.27]
	R3	3.42 [1.26; 4.94]	3.39 [1.25; 4.97]	3.55 [1.06; 5.40]	4.10 [1.22; 6.01]
lBF	R1	8.74 [5.49; 16.24]	9.41 [6.09; 15.62]	8.49 [4.89; 11.53]	8.98 [5.13; 11.86]
	R2	3.72 [2.27; 9.33]	3.87 [2.34; 8.77]	3.10 [2.36; 5.95]	3.34 [2.51; 6.43]
	R3	3.11 [1.80; 3.85]	3.02 [1.86; 3.77]	2.55 [1.89; 3.23]	2.65 [1.97; 3.32]
rST	R1	6.50 [4.13; 9.83]	6.51 [4.16; 10.29]	5.91 [3.90; 10.84]	7.03 [4.86; 13.04]
	R2	3.31 [2.21; 3.99]	3.28 [2.30; 4.03]	2.76 [2.27; 5.70]	3.17 [2.63; 6.10]
	R3	2.84 [1.49; 3.29]	2.82 [1.53; 3.22]	2.26 [1.67; 3.64]	2.65 [1.72; 4.69]
lST	R1	6.22 [3.74; 14.96]	6.59 [3.50; 17.03]	7.42 [3.48; 13.38]	8.19 [3.74; 14.67]
	R2	3.46 [2.29; 7.08]	3.50 [2.27; 7.29]	3.99 [2.79; 6.16]	4.50 [3.15; 7.27]
	R3	3.07 [2.19; 7.40]	2.86 [2.15; 7.23]	3.81 [2.12; 4.42]	4.29 [2.31; 4.89]

Abbreviations: aMVIC, average of maximum voluntary isometric contraction; aP, average of maximums of each repetition; Ch, channel; lBF, left biceps femoris; lST, left semitendinosus; mMVIC, maximum value of maximum voluntary isometric contraction; mP, maximum of each repetition; R, range; rBF, right biceps femoris; rST, right semitendinosus. Note: Data are presented as medians and percentiles [25%; 75%].

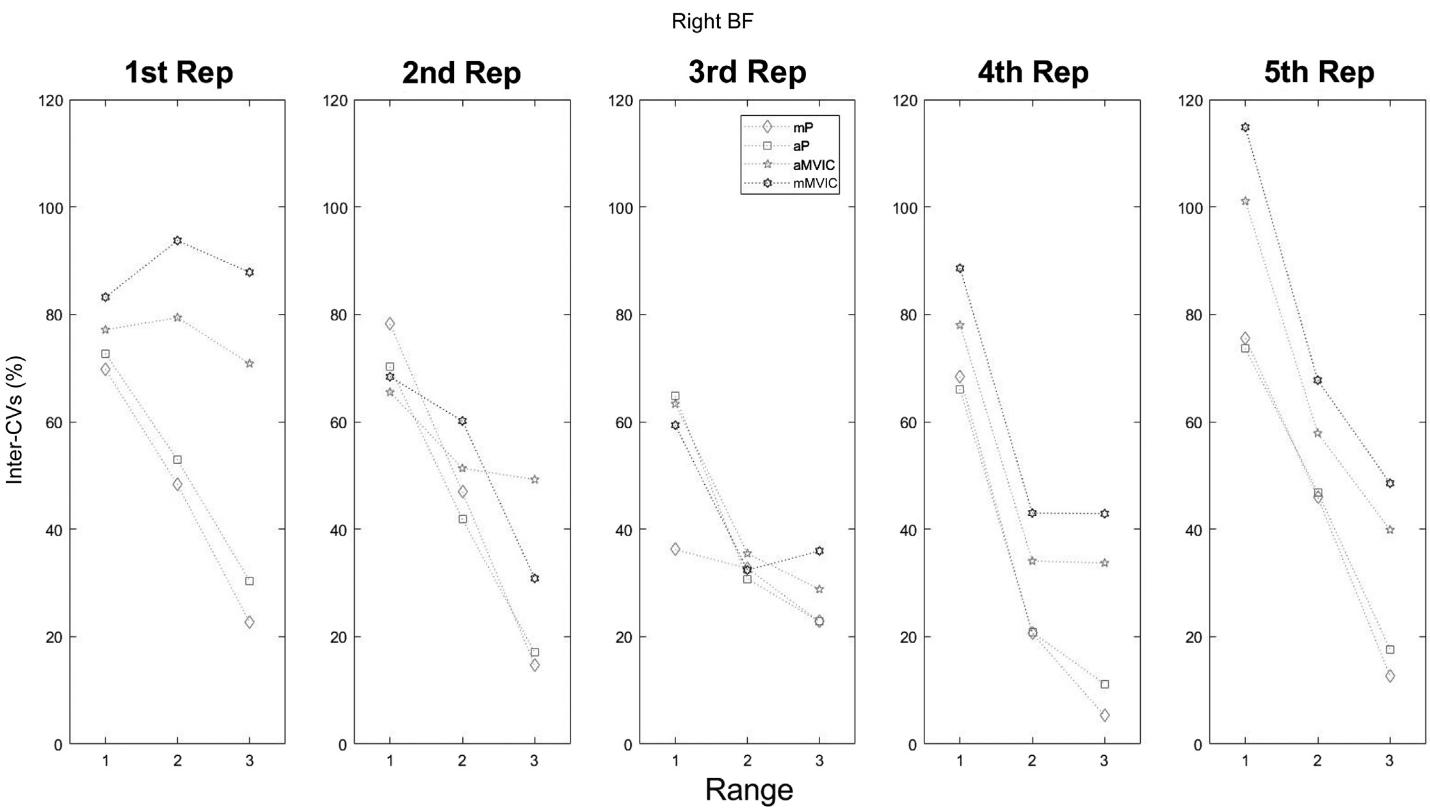

Figure 1 — Inter-CVs by range (R1: 0%–33%, R2: 33%–66%, and R3: 66%–100%) and by Rep according to each normalization method in right BF. aMVIC indicates average of maximum voluntary isometric contraction; aP, average of maximums of each repetition; BF, biceps femoris; inter-CVs, intersubject variation coefficients; mMVIC, maximum value of maximum voluntary isometric contraction; mP, maximum of each repetition; Rep, repetition.

normalization values extracted from the MVIC test. Despite the relevance of eccentric contractions for sports performance and muscle injury prevention and rehabilitation, there are few studies that investigate the variations associated with electromyogram normalization in this type of contractions.

Burden[15] suggests that normalization methods using parameters taken directly from the task being performed should be used to reduce interindividual variability.[15] In these contexts, Ditroilo et al[13] performed a kinematic and electromyographic description of the NHE using a normalizing parameter from a maximum

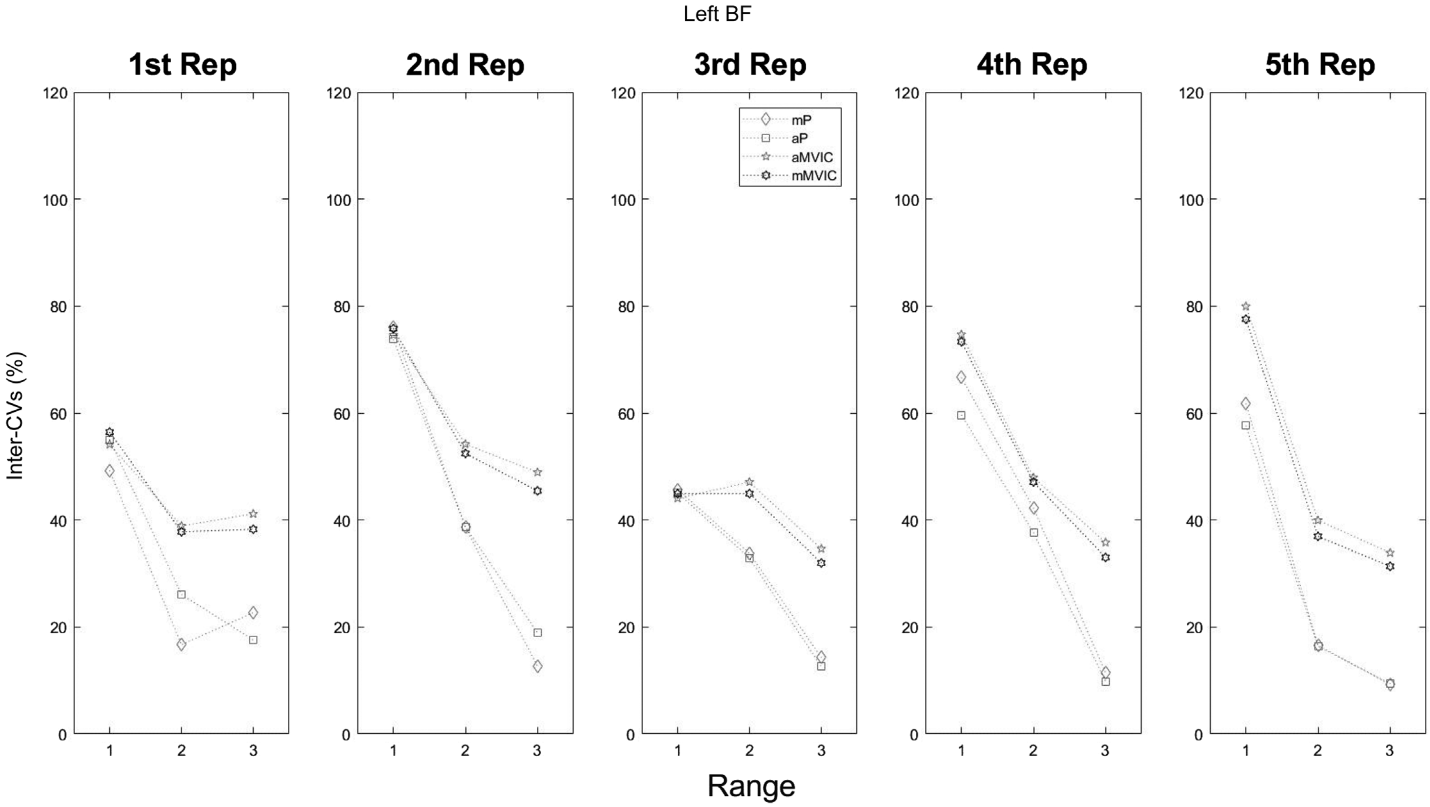

Figure 2 — Inter-CVs by range (R1: 0%–33%, R2: 33%–66%, and R3: 66%–100%) and by Rep according to each normalization method in left BF. aMVIC indicates average of maximum voluntary isometric contraction; aP, average of maximums of each repetition; BF, biceps femoris; inter-CVs, intersubject variation coefficients; mMVIC, maximum value of maximum voluntary isometric contraction; mP, maximum of each repetition; Rep, repetition.

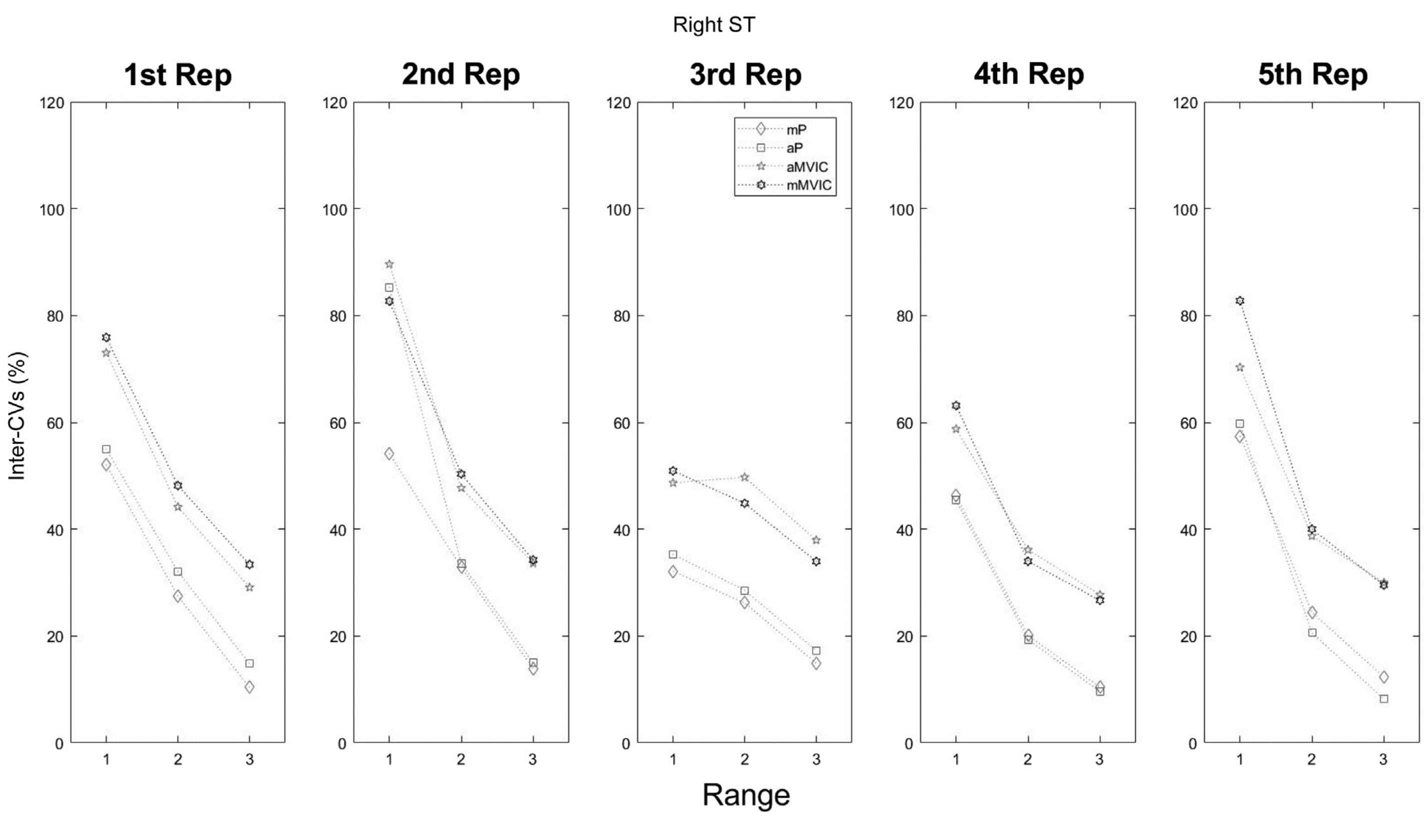

Figure 3 — Inter-CVs by range (R1: 0%–33%, R2: 33%–66%, and R3: 66%–100%) and by Rep according to each normalization method in right ST. aMVIC indicates average of maximum voluntary isometric contraction; aP, average of maximums of each repetition; inter-CVs, intersubject variation coefficients; mMVIC, maximum value of maximum voluntary isometric contraction; mP, maximum of each repetition; Rep, repetition; ST, semitendinosus.

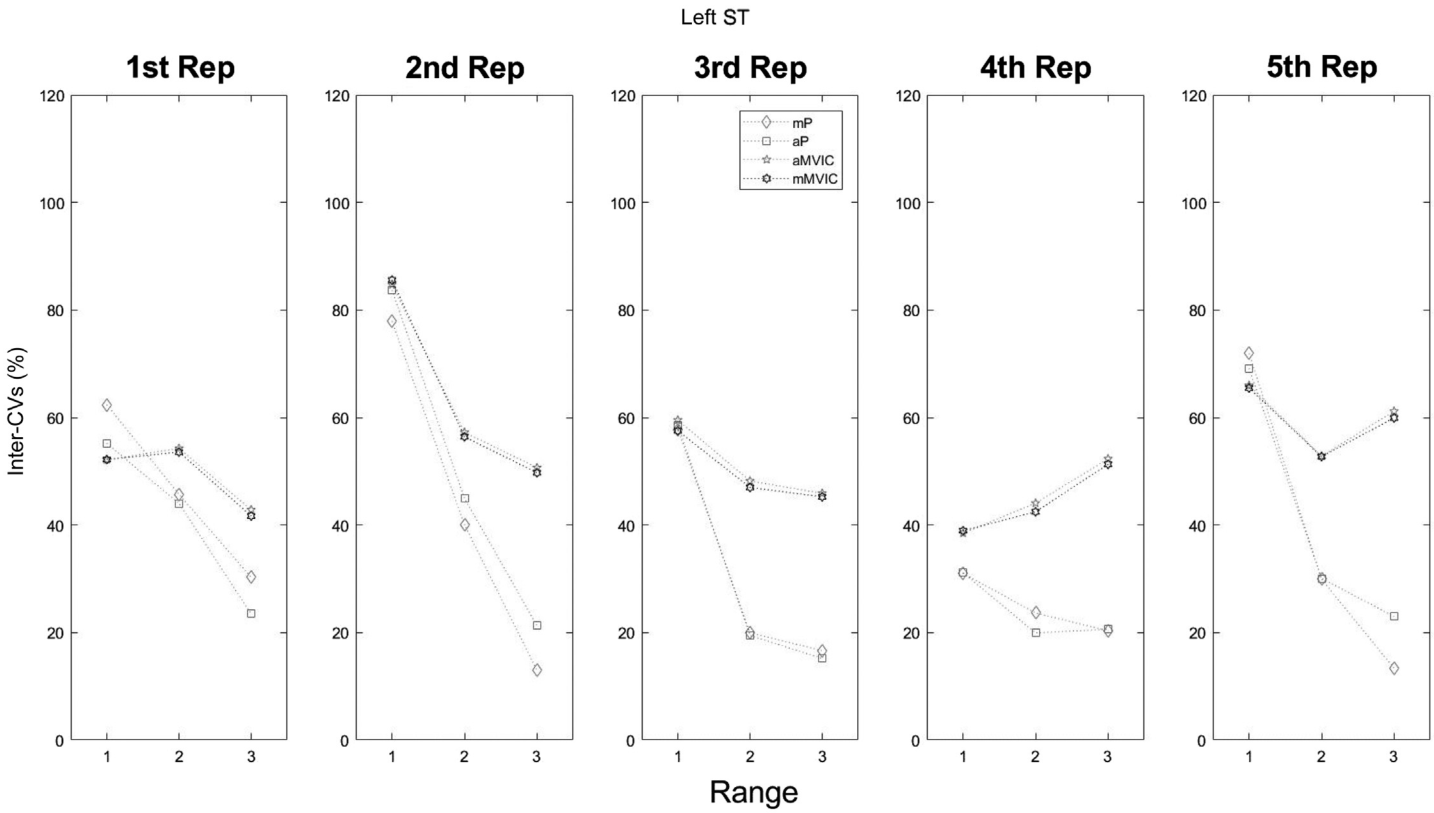

Figure 4 — Inter-CVs by range (R1: 0%–33%, R2: 33%–66%, and R3: 66%–100%) and by Rep according to each normalization method in left ST. aMVIC indicates average of maximum voluntary isometric contraction; aP, average of maximums of each repetition; inter-CVs, intersubject variation coefficients; mMVIC, maximum value of maximum voluntary isometric contraction; mP, maximum of each repetition; Rep, repetition; ST, semitendinosus.

peak of EMG hamstring eccentric activity reached in 1 repetition.[13] Moreover, intrasession coefficients of variation were used as a measure of electromyography reliability, with values ranging between 18.02% and 21.99%. The highest values of coefficients of variation were reached in the first degrees of movement of each repetition, which is consistent with our research. In another study evaluating variations of NHE using mMVIC as normalization method, the authors found a high variability in hamstring electromyogram values when comparing their results with similar studies, suggesting that the variability observed in the mentioned muscles could be associated with the normalization procedure.[16]

Bolgla and Uhl[17] analyzed the reliability of the MVIC (ie, mMVIC in our research), mean dynamic activity (aP), and peak dynamic activity (mP) as normalization procedures in hip muscles.[17] In their research, the highest values of inter-CVs were recorded when using the mMVIC method (55%–77%) followed by aP (26%–61%) and mP (19%–44%), concluding that parameters from the task achieve lower variability in electromyograms, which is consistent with our research. Similarly, they reported intra-CVs from 11% to 22% for all normalization methods, also in agreement with our results.

While the normalization to mean/peak from an EMG recording reduced the intersubject amplitude variation, this procedure has some limitations. This method normalizes signals to their peak/mean, limiting comparison between muscles that have different levels of activations during a task (ie, NHE). The appropriateness of a normalization method for a specific context is not straightforward according to a recent consensus for experimental design in electromyography.[1] Normalization to peak/mean from the task can be useful to compare EMG amplitude within a subject or single muscle between conditions (ie, a treatment or training program), but caution must be taken when comparing tasks or the same muscle activation in different participants because EMG amplitude differences are intentionally removed by the normalization method. Future research could describe whether different normalization methods have an influence on electromyographic activity variables when comparing study subjects in dynamic conditions.

We must state some limitations of our study. As our sample corresponds to asymptomatic male rugby players, the results should be carefully translated to injured or other sports populations. Moreover, participants had experience performing the NHE in their discipline's training. Regarding the normalization methods, there was a limitation associated with the recording of the MVIC, as it is assumed that such contraction comes from maximal effort; therefore, the participants were verbally encouraged to give their maximum effort during the execution of the test.

Conclusion

We can conclude that the methods that allow less variability in sEMG amplitude normalization in eccentric hamstring contractions are those that use parameters extracted from the execution of the task.

Acknowledgment

This research was promoted by the Nucleus of Research in Rehabilitation and Human Movement Sciences (Decree 1076/2015).

References

1. Besomi M, Hodges PW, Clancy EA, et al. Consensus for experimental design in electromyography (CEDE) project: amplitude normalization matrix. *J Electromyogr Kinesiol*. 2020;53:102438. PubMed ID: 32569878 doi:10.1016/j.jelekin.2020.102438
2. Farina D. Interpretation of the surface electromyogram in dynamic contractions. *Exer Sport Sci Reviews*. 2006;34(3):121–127. PubMed ID: 16829739 doi:10.1249/00003677-200607000-00006
3. Lehman GJ, McGill SM. The importance of normalization in the interpretation of surface electromyography: a proof of principle. *J Manipulative Physiol Ther*. 1999;22(7):444–446. PubMed ID: 10519560 doi:10.1016/S0161-4754(99)70032-1
4. Del Vecchio A, Bazzucchi I, Felici F. Variability of estimates of muscle fiber conduction velocity and surface EMG amplitude across subjects and processing intervals. *J Electromyogr Kinesiol*. 2018;40:102–109. PubMed ID: 29705496 doi:10.1016/j.jelekin.2018.04.010
5. Tabard-Fougère A, Rose-Dulcina K, Pittet V, Dayer R, Vuillerme N, Armand S. EMG normalization method based on grade 3 of manual muscle testing: within- and between-day reliability of normalization tasks and application to gait analysis. *Gait Posture*. 2018;60:6–12. PubMed ID: 29121510 doi:10.1016/j.gaitpost.2017.10.026
6. Tsaklis P, Malliaropoulos N, Mendiguchia J, Korakakis V, Tsapralis K, Pyne D, Malliaras P. Muscle and intensity based hamstring exercise classification in elite female track and field athletes: implications for exercise selection during rehabilitation. *Open Access J Sports Med*. 2015;6:209–217. PubMed ID: 26170726 doi:10.2147/oajsm.s79189
7. Ball S, Halaki M, Orr R. Training volume and soft tissue injury in professional and non-professional rugby union players: a systematic review. *Br J Sports Med*. 2017;51:1012–1020. PubMed ID: 28525884 doi:10.1136/bjsports-2015-095926
8. Hermens HJ, Freriks B, Disselhorst-Klug C, Rau G. Development of recommendations for SEMG sensors and sensor placement procedures. *J Electromyogr Kinesiol.* 2000;10:361–374.
9. Stratford PW, Balsor BE. A comparison of make and break tests using a hand-held dynamometer and the Kin-Com. *JJ Orthop Sports Phys Ther*. 1994 Jan;19(1):28–32. PubMed ID: 8156061 doi:10.2519/jospt.1994.19.1.28
10. Cronin NJ, Kumpulainen S, Joutjärvi T, Finni T, Piitulainen H. Piitulainen. Spatial variability of muscle activity during human walking: The effects of different EMG normalization approaches. *Neurosci.* 2015;300:19–20. doi:10.1016/j.neuroscience.2015.05.003
11. Guruhan S, Kafa N, Ecemis ZB, Guzel NA. Muscle Activation Differences During Eccentric Hamstring Exercises. *Sports Health*. 2021 Mar;13(2):181–186. doi:10.1177/1941738120938649
12. Josephson MD, Knight CA. Comparison of neural excitation measures from the surface electromyogram during rate-dependent muscle contractions. *J Electromyogr Kinesiol*. 2019;44:15–20. PubMed ID: 30465942 doi:10.1016/j.jelekin.2018.11.004
13. Ditroilo M, De Vito G, Delahunt E. Kinematic and electromyographic analysis of the Nordic hamstring exercise. *J Electromyogr Kinesiol*. 2013;23(5):1111–1118. PubMed ID: 23809430 doi:10.1016/j.jelekin.2013.05.008
14. Chuang TD, Acker SM. Comparing functional dynamic normalization methods to maximal voluntary isometric contractions for lower limb EMG from walking, cycling and running. *J Electromyogr Kinesiol*. 2019;44:86–93. PubMed ID: 30551007 doi:10.1016/j.jelekin.2018.11.014
15. Burden A. How should we normalize electromyograms obtained from healthy participants? What we have learned from over 25 years of research. *J Electromyogr Kinesiol*. 2010;20(6):1023–1035. PubMed ID: 20702112 doi:10.1016/j.jelekin.2010.07.004
16. Sarabon N, Marusic J, Markovic G, Koznic Z. Kinematic and electromyographic analysis of variations in Nordic hamstring exercise. *PLoS One*. 2019;14(10):1–16. PubMed ID: 31644582 doi:10.1016/j.jelekin.2013.05.008
17. Bolgla LA, Uhl TL. Reliability of electromyographic normalization methods for evaluating the hip musculature. *J Electromyogr Kinesiol*. 2007;17(1):102–111. PubMed ID: 16423539 doi:10.1016/j.jelekin.2005.11.007